E-commerce Sales Performance Analysis

Reporting Period: January 2024 – December 2025

Tools: SQL Server (SSMS), Excel

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1. Executive Summary

The e-commerce business experienced a decline in sales performance in 2025 compared with 2024. Total revenue decreased from 35,838 to 27,764, representing a reported decline of 22.5%. Over the same period, the number of orders fell from 118 to 102, while newly acquired customers decreased from 49 to 11.

The analysis suggests that the revenue decline was accompanied by fewer orders, fewer items sold, and a lower average order value. At product level, Footwear and Electronics emerged as the main areas of concern, while Clothing was the only category to record revenue growth.

These findings point to two priorities for the business: understanding the decline in customer acquisition and investigating the weaker performance of key product categories.

However, the analysis also identified figures that require further validation. The category-level revenue totals do not fully reconcile with the reported annual revenue, so the figures should be verified before final publication.

2. Business Problem

The business needs to understand not only how much revenue it generates, but also what drives changes in sales performance over time.

The purpose of this analysis was to assess the e-commerce business's performance across 2024 and 2025, identify areas of growth and decline, and investigate which customer and product trends could help explain the change in revenue.

The central business question was:

What contributed to the decline in revenue in 2025, and which areas should the business prioritise to improve sales performance?

Rather than treating the revenue decline as an isolated result, the analysis examined customer acquisition, order volume, average order value, product categories, individual product sales, and unit prices.

3. Business Questions

The analysis focused on the following questions:

1. How did revenue and customer acquisition change between 2024 and 2025?
2. Did the number of orders and average order value change?
3. Which product categories contributed most to the revenue decline?
4. Which individual products experienced the largest decreases in units sold?
5. Were changes in average unit prices associated with the decline in sales?
6. What actions could the business take to address the identified performance issues?

4. Methodology

The analysis used the Customers, Orders, Order Items, and Products tables to investigate customer and sales performance.

SQL queries were used to aggregate revenue, count orders, calculate average order value, compare units sold across years, and examine product-level performance.

The following calculations were applied:

- **Revenue:** Sum of quantity multiplied by unit price.
- **Customer acquisition:** Number of customers grouped by their signup year.
- **Order volume:** Number of distinct orders placed in each year.
- **Average order value (AOV):** Total revenue divided by the number of distinct orders.
- **Product performance:** Comparison of revenue and units sold by category and individual product.
- **Average unit price:** Average recorded unit price for each product in each year.

The analysis compared the two complete calendar years, 2024 and 2025. Product categories and individual products were then examined to identify the area's most closely associated with the overall decline.

5. Key Findings and Business Implications

5.1 Revenue declined alongside customer acquisition

Reported revenue decreased from 35,838 in 2024 to 27,764 in 2025, a decline of approximately 22.5%.

Customer acquisition also weakened considerably. The number of newly acquired customers fell from 49 to 11, representing a decrease of approximately 77.6%.

This is a significant finding because the business acquired substantially fewer customers in 2025 while also generating less revenue.

Business implication: The reduction in new customers raises concerns about the business's ability to attract new buyers. However, the available results do not establish whether lower acquisition caused the revenue decline. Further investigation would be needed to examine traffic, marketing activity, conversion rates, and the purchasing behaviour of existing customers.

5.2 Fewer orders and lower average order value accompanied the revenue decline

The total number of orders decreased from 118 in 2024 to 102 in 2025, a decline of approximately 13.6%.

The separate average order value calculation returned 303.71 for 2024 and 278.16 for 2025, suggesting a decline of approximately 8.4%.

Taken together, these figures suggest that the business generated fewer orders and that the average value of an order also decreased.

Business implication: The business may need to address both purchase frequency and the value of each transaction, rather than focusing exclusively on attracting new customers.

These findings require a final consistency check: the reported revenue, order counts, and AOV figures do not currently reconcile exactly. The final report should use verified totals from the same underlying dataset and calculation.

5.3 Electronics and Footwear were the main category-level concerns

The category-level analysis identified Electronics and Footwear as the largest contributors to the observed revenue decline.

Product category	Revenue 2024	Revenue 2025	Approximate change
Electronics	21,827.29	16,797.82	-23.1%
Footwear	5,249.30	2,774.63	-47.1%
Home office	1,619.46	1,259.58	-22.2%
Sports	1,139.43	899.55	-21.1%
Accessories	2,049.14	1,884.21	-8.0%
Bags	2,049.59	1,949.61	-4.9%
Clothing	1,904.58	2,429.46	+27.6%

Electronics recorded the largest absolute revenue decline, decreasing by approximately 5,029. Footwear followed, with revenue decreasing by approximately 2,475.

Clothing was the only category to record revenue growth, increasing by approximately 27.6%.

Business implication: Electronics and Footwear deserve priority in further investigation because of their contribution to the decline. Clothing provides a useful comparison case: the business should examine what is working in this category before deciding whether successful approaches can be applied elsewhere.

5.4 Lower sales volumes help explain the category declines

The analysis compared the number of items sold in each category.

Electronics units sold decreased from 271 in 2024 to 218 in 2025, a decline of approximately 19.6%.

Footwear units sold fell from 70 to 37, a decline of approximately 47.1%.

Other categories also recorded lower sales volumes, including home office, Sports, Accessories, and Bags. Clothing moved in the opposite direction, with units sold increasing from 42 to 54.

These results are consistent with the revenue patterns: categories selling fewer items generally experienced lower revenue, while Clothing recorded growth in both units sold and revenue.

Business implication: The results point towards lower sales volumes as an important factor in the revenue decline. However, volume alone does not explain why customers purchased fewer items. The business would need additional information to determine whether the pattern reflects weaker demand, lower website traffic, stock availability, product preferences, or other factors.

5.5 Product-level analysis identified specific areas of concern

To investigate the Electronics and Footwear declines further, the analysis examined units sold for individual products.

Product	Units sold in 2024	Units sold in 2025	Change
Running Shoes	70	37	-47.1%
Smart Watch	67	44	-34.3%
Wireless Headphones	28	31	+10.7%

Running Shoes experienced the steepest decline in units sold, falling by 47.1%. Smart Watches also experienced a considerable decline of approximately 34.3%.

Wireless Headphones were the only product in this selection to record an increase in units sold, rising at approximately 10.7%.

Business implication: The product-level results give the business a more specific starting point for action. Rather than responding to the overall decline with a broad sales initiative, the business can prioritise investigation into Running Shoes and Smart Watches while examining the factors behind Wireless Headphones' growth.

The results identify where performance weakened, but they do not establish why it happened. Stock availability, customer demand, competition, marketing exposure, and product reviews would be useful areas for further investigation.

5.6 Average unit prices remained unchanged

The analysis compared the average recorded unit price of the six products across both years.

No change was observed in the average recorded unit prices for these products.

This supports the interpretation that lower quantities sold, rather than changes in average transaction prices for these products, were associated with their revenue declines.

Business implication: The business should not assume that discounting or price reductions are necessary to improve performance. Before changing prices, it would be more useful to investigate demand, product availability, marketing reach, and customer feedback.

The finding applies to the six products examined and does not establish that prices remained unchanged across the entire product catalogue.

6. Recommendations

The findings suggest four priorities for the business.

6.1 Investigate the decline in customer acquisition

New customer acquisition decreased by approximately 77.6%, making it one of the most significant changes identified.

The business should examine website traffic, marketing campaigns, customer conversion rates, and acquisition channels to identify where potential customers may be dropping out.

This would help determine whether the decline reflects reduced exposure to potential customers or a lower likelihood of converting visitors into buyers.

6.2 Prioritise Electronics and Footwear

Electronics and Footwear experienced the largest category-level revenue declines.

The business should investigate product availability, customer demand, marketing exposure, product reviews, and competitor offerings in these categories. Running Shoes and Smart Watches should receive particular attention because of their substantial decreases in units sold.

Any promotional activity should be based on the findings of that investigation rather than introduced automatically.

6.3 Learn from Clothing and Wireless Headphones

Clothing recorded revenue growth of approximately 27.6%, while Wireless Headphones recorded a 10.7% increase in units sold.

The business should examine whether these products benefited from stronger demand, better visibility, greater availability, or other favourable conditions.

Understanding the differences may help identify practices that could be tested in weaker-performing categories.

6.4 Monitor sales performance consistently

The business should monitor revenue, order volume, average order value, customer acquisition, and units sold by category and product monthly.

A monthly view would help identify when performance began to deteriorate, rather than only revealing the difference at the end of the year.

Tracking these metrics together would also help the business assess whether future actions are improving sales performance.

7. Limitations and Further Investigation

Although the analysis identified several important patterns, some limitations should be acknowledged.

Revenue reconciliation: The category-level revenue figures for 2025 total approximately 27,994.86, which differs from the reported annual revenue of 27,764. These figures must be reconciled before the final report is published.

Order-count validation: The customer-segment query initially used a non-distinct order count after joining Orders to Order Items. This can count the same order multiple times when an order contains multiple items. Customer-segment order counts should therefore be recalculated using distinct order IDs before being reported.

Average order value validation: The reported annual AOV values should be recalculated using the reconciled revenue totals and distinct order counts to ensure consistency.

Limited explanation of causes: The dataset identifies where performance declined but does not, by itself, establish why. Additional information on website traffic, marketing, stock availability, customer feedback, and competition would be required to determine the underlying causes.

Limited pricing scope: The unchanged prices were observed for six selected products, not the entire catalogue. These findings should not be generalised to all products without further analysis.

Reporting period: The analysis covers 2024 and 2025. It identifies the difference between those years but does not establish whether the decline continued, began earlier, or reflects a longer-term trend.

8. Conclusion

The analysis found that the e-commerce business performed less strongly in 2025 than in 2024. Reported revenue and new customer acquisition declined, while order volume and the average order value calculation also indicated weaker performance.

The investigation moved beyond the headline revenue figure to identify Electronics and Footwear as the main category-level concerns. At product level, Running Shoes and Smart Watches experienced particularly substantial declines in units sold. Meanwhile, Clothing and Wireless Headphones provided examples of growth within an otherwise weaker sales environment.

The findings suggest that the business should prioritise understanding the decline in customer acquisition, investigate weaker-performing products, and learn from areas that continued to grow. However, the reasons behind the declines remain hypotheses until supported by further evidence.

Before drawing final conclusions, the revenue totals, order counts, and AOV figures must be reconciled. Once validated, these findings can provide a stronger foundation for targeted business decisions and ongoing performance monitoring.

Ultimately, the value of this analysis is not simply identifying that revenue decreased. It is narrowing the problem down to specific areas of the business and identifying where further investigation and action are most likely to be useful.